

David Corcoran

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EDUCATION

Georgetown University

Graduation Date: Dec 2025

Master of Science in Data Science and Analytics

Virginia Tech

Graduation Date: May 2022

Bachelor of Science in Business Information Technology - *Computer Decision Support Systems Major*

Bachelor of Science in Finance - *Investment Management and Chartered Financial Analyst Major*

WORK EXPERIENCE

Data and Analytics Intern | NASA

June 2025 – Aug 2025

- Supported the Aerosciences Evaluation and Test Capabilities (AETC) team, responsible for managing NASA's wind tunnels.
- Designed and executed a capital budgeting methodology to evaluate whether to maintain or rebuild a portfolio of 12 wind tunnels.
- Developed cost forecasting models using polynomial regression and non-linear curve fitting techniques to estimate future operational and maintenance expenses and inform Net Present Cost (NPC) analysis.
- Ran Monte Carlo simulations to quantify uncertainty and variability in cost projections and decision outcomes.
- Performed a sensitivity analysis to evaluate the impact of planning assumptions on investment decisions and prioritization.

Federal Technology Risk Management Associate | KPMG

July 2022 – July 2024

- Obtained a DoD Secret Clearance to support sensitive government projects.
- Conducted on-site visits across the US, collaborating with clients to evaluate warehouse IT controls and identify control gaps.
- Authored a Corrective Action Plan, containing an E2E process walkthrough document, risk and control matrix, control gap analysis, and responsibility-accountability matrix to address identified IT process deficiencies.
- Supported the development of a program governance charter, conducted a current-state SoD/ICAM assessment, and created a Microsoft Power BI dashboard to facilitate the enterprise-wide ICAM implementation.

PROJECTS

"Off the Beaten Path" Travel Recommender, Machine Learning and Application Deployment (DSAN 6700)

- Developed a full-stack travel recommendation system surfacing lesser-known destinations using hybrid information retrieval, combining BM25 keyword search and FAISS dense embeddings (modernBERT) for relevance scoring.
- Built a scalable data ingestion pipeline to crawl WordPress blogs and Reddit travel posts, extracting destinations with spaCy NER, sentence embeddings, and geo-resolution using geopy and OpenStreetMap APIs.
- Deployed a Docker-containerized microservices architecture with a FastAPI backend on Render, HuggingFace Spaces frontend UI using Streamlit, PostgreSQL metadata store, Redis caching layer, and AWS S3 object storage for media and embeddings.

Deep Reinforcement Learning for Soccer On-Ball Decision Making, Reinforcement Learning (DSAN 6650)

- Implemented offline reinforcement learning through Implicit Q-Learning (IQL) to optimize real-time soccer decisions (pass, shoot, dribble) on 2+ seasons of EPL event data (~2,900 events per match from Impect).
- Engineered multi-dimensional state representations (35 features $\times$ 15-30 event sequences) incorporating pitch coordinates, pressure metrics, game phase, player FIFA ratings, and temporal context to capture tactical dynamics.
- Designed a custom Gym environment with a hierarchical reward structure and discrete action space mapping to field grid zones.

Vehicle Collision Neural Network Classification Models, Neural Networks and Deep Learning (DSAN 6600)

- Processed ~60,000 image frames from 1,500 annotated dashcam videos, utilizing YOLOv8 to obtain object-level spatial features.
- Created LSTM, GRU, and Transformer-based sequence models to classify vehicle collisions from real-world dashcam footage into no collision/collision categories based on temporal patterns in object features.
- Achieved the highest performance using a GRU-based architecture, with 68% accuracy and a 69% F1 score, highlighting the ability for RNN models to effectively learn temporal patterns from raw video sequences and object-level features.

SKILLS

Programming: Python, R, SQL, JavaScript, HTML, CSS, PHP, VBA, VB.NET, Selenium, Bootstrap

Data Analysis & Visualization: Pandas, NumPy, SciPy, Matplotlib, Seaborn, NetworkX, Plotly, Closeroad, Microsoft Power BI

Machine Learning: PyTorch, Keras, TensorFlow, scikit-learn, RLlib, MLFlow

Miscellaneous: Docker, PySpark, AWS EC2 and S3, Github, Git, bash, Quarto, Jupyter Notebooks, Microsoft Access